

The eXtreme Gradient Boosting Method Optimized by Hybridized Sine Cosine Metaheuristics for Ship Vessel Classification



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Abstract Ship classification is essential in coastal areas to ensure safety, protect the environment and improve maritime security. It also allows the optimization of resource allocation and can boost economic growth. Therefore, vessel identification is crucial to employ appropriate security measures. However, communication interruptions can happen during poor weather conditions, which could hinder the overall safety of vessels in the area. Security is, therefore, a main pivotal aspect that drives forward the vessel identification systems. This paper tackles this problem by proposing an XGBoost machine learning model that is optimized by an enhanced variant of the sine cosine metaheuristic algorithm that has the role of identifying and classifying naval vessels. The proposed method has been compared to the results attained by other cutting-edge metaheuristics algorithms, and experimental outcomes show that it obtained supreme results for this particular task.

Keywords Ship classification · Machine learning · XGBoost · Metaheuristics optimization · Sine cosine algorithm

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1 Introduction

Ship classification is vital in coastal areas, as it enables effective regulation, monitoring, and management of ships operating in these areas, leading to sustainable coastal development and the preservation of marine ecosystems. Accurate classification of ships with respect to their type, dimensions, and capabilities can help the coastal authorities to enforce the appropriate regulations and standards to ensure safe navigation and preserve the ecologically sensitive environment by minimizing pollution and environmental impact. Additionally, this classification can help in detecting potential security threats (indicating illegal activities such as smugglers or unauthorized entry) and also support the ports to allocate appropriate facilities based on the ship category, therefore avoiding congestion and improving operational efficacy [2, 41].

Ships typically use Automatic Identification System (AIS) to improve security [31], which electronically sends data about the vessel to other ships and coastal control stations. It combines data from Global Positioning System (GPS) with onboard sensors and digital radio communication to allow exchanging data. However, since AIS relies on a radio-based system (VHF), in poor weather conditions, communication interruptions, and interference can occur. Therefore, a robust system to classify different types of vessels is necessary to support the authorities in enforcing appropriate measures to keep the coastal areas safe.

The state-of-the-art machine learning algorithms are capable of very accurate classifications and even act proactively to make sure that appropriate preventive activities are performed in case of suspicious behavior. They can be applied to very intricate non-linear problems, delivering excellent results even without explicit programming. The extreme gradient boosting (XGBoost) methodology is widely acknowledged as an exceptional for its remarkable performance in diverse domains. It relies on a gradient boosting framework, improving the model's performance while reducing both bias and variance at the same time. Due to its close association with tree-based methods, it is often classified as such. To tune the intricate hyperparameters of XGBoost, an optimizer must be employed for each particular classification task. Metaheuristic algorithms are well-known for their ability to produce high-performance results. These algorithms shine in tackling problems categorized as nondeterministic polynomial time-hard (NP-hard). To tune the XGBoost hyperparameters, the authors have chosen an enhanced version of the famous sine cosine algorithm (SCA) [26], that has shown exquisite results when dealing with NP-hard optimization tasks.

This paper contains several important contributions to this domain. It proposes a robust classification system designated to identify different types of naval vessels based on the XGBoost model. To tune the hyperparameters of the XGBoost, this research suggests an enhanced variant of the SCA algorithm that achieves supreme performance in comparison to the existing methods. Finally, a comprehensive comparative analysis was conducted on a benchmark real-world AIS dataset, providing an important use case to evaluate the suggested methodology for the multiclass classification challenge. The main contributions are listed below:

- Proposed enhancements to the elementary SCA algorithm.
- Development of a robust classification system based on the XGBoost model, tuned by the suggested improved SCA metaheuristics.
- Extensive testing was executed over a benchmark AIS dataset, where the performance of seven XGBoost models that utilize various metaheuristics was evaluated and compared.

The rest of this document structured as such: Sect. 2 brings forward the important background information that relates to the utilized methods, together with the comprehensive literature survey. Section 3 firstly introduces the basic SCA algorithm, followed by the proposed modified algorithm. Section 4 describes the dataset utilized in the simulations, provides information about the experimental setup and brings forward the simulation outcomes and discussion of the attained results. In the end, Sect. 5 wraps up the study and provides final observations.

2 Background and Related Works

An review of the many methodologies used to classify naval vessels, followed by a brief history of the XGBoost model used in the article are presented are provided in this section. Finally, a brief overview of metaheuristic optimization methods and their practical applications in many disciplines is offered.

2.1 Vessel Classification

Machine learning approaches were frequently employed to help in ship classifying tasks. The authors in [33] tested various methods to identify potential candidates prone to accidents in poor weather conditions (hurricane season). By proper identification of ships in trouble, marine safety can be leveraged. The authors determined a support vector machine (SVM) to be the most suitable candidate for this task.

Paper [23] explored the potential of determining the class of vessel only by using trajectory data obtained from AIS messages and was able to successfully distinguish among five classes of ships. The authors in [30] proposed a deep learning framework revolving around recurrent neural networks to handle historical AIS data to allow prediction of future vessel behavior and improving maritime safety. Convolutional neural networks were used in [34] for ship detection based on the AIS data merged with radar images, with respectable results.

Deep learning approaches were used to classify the vessel trajectory [15] and for route planning [24] based on the AIS data as well, with promising results. Despite the decent amount of publications that use some of the artificial intelligence approaches to deal with vessel classification for different purposes, it can be noted that there is a

significant gap with respect to the metaheuristics-based tuning of the models, which could potentially yield even better results in this domain.

2.2 XGBoost Overview

The XGBoost model, a contemporary and widely utilized machine learning method, has gained recognition for its remarkable power and accuracy [12, 13]. To minimize the loss function, optimizing the model's parameters is of utmost importance since they greatly affect the model's behavior. Through a combination of several weaker models, XGBoost builds a robust model that yields more precise forecasts. XGBoost attains exceptional outcomes by incorporating an approach such as gradient boosting, regularization, and control value tuning. This model works on the principle of accurately forecasting results by leveraging adopted patterns of entangled dependencies that exist among the input data and the target variable.

In common with many other recent machine learning models, the XGBoost model provides a collection of particular hyperparameters that demand proper adaptation to demonstrated a desired quality of results. The choice and manipulation of these hyperparameters directly impact the configuration of different models' properties with respect to the particular problem at hand. XGBoost offers a mechanism for meticulous parameter adjustment, enabling high precision governance of learning phase. The overarching objective is to establish a balance to achieve consistency over indicators, including accuracy, speed, and the XGBoost generalization ability. To attain supreme and optimal outcomes, an iterative procedure of parameter optimization is frequently necessary that involves appropriate data validation that depends on the previous evaluations. The fitness function that defines the XGBoost behavior is provided in Eq. (1) [12]:

$$\text{obj}(\Theta) = L(\theta) + \Omega(\Theta), \quad (1)$$

In XGBoost, the provided formula reflects the sum loss and regularization operators. Θ denotes the collection of XGBoost hyperparameters, and $L(\theta)$ denotes the loss function. Furthermore, $\Omega(\Theta)$ denotes the regularization term used to control the XGBoost structure complexity. The mean squared error is formally defined as the loss function.

$$L(\Theta) = \sum_i (y_i - \hat{y}_i)^2, \quad (2)$$

where y_i is representing the estimated value, and $\hat{y}_i$ refers to the predicted values for the target variable relative to every i iteration.

$$L(\Theta) = \sum_i [y_i \ln(1 + e^{-\hat{y}_i}) + (1 - y_i) \ln(1 + e^{\hat{y}_i})], \quad (3)$$

The goal of this procedure is to calculate observed and expected value discrepancy, specifically between actual and anticipated values. This function's goal and conclusion is to minimize the total value of the loss function, hence improving classification results.

2.3 *Metaheuristics Optimization*

In computational problems, stochastic approaches are frequently required because the NP-hard nature of the problems makes deterministic methods impractical. Several branches of metaheuristics can be categorized based the sources of their inspiration, such as swarm behavior of insects or evolution [16, 17, 36]. Among these families, swarm intelligence algorithms have gained significant influence.

The presence of such a large scope of metaheuristic solutions may be linked to the “no free lunch theorem” [38], which suggests that different problems require the exploration of unique solutions. There is no universal best algorithm that can effectively solve every optimization problem in existence. Therefore, a method performing exceptionally well for one task can entirely fail for another problem.

In recent times, there has been a significant growth in the utilization of metaheuristics approaches for real-world applications. The application domains cover optimization of the wireless sensor networks and IoT networks [43, 45, 48], processing and classification of MRI images [10, 42], credit card deceits detection [18, 32], prediction of the COVID-19 cases [44, 47], enhancement of the fog, cloud, and cloud-edge computing networks [11, 46], intrusion detection systems [1, 8, 35], feature selection task [3, 9], green energy production and consumption levels prediction [4, 7, 37], optimization of a wide variety of ML structures [6, 20], and finally air pollution estimation and monitoring of the environment [5, 19, 21].

3 **Proposed Method**

In this section, we begin by presenting the fundamental principles of the original SCA metaheuristics algorithm. Subsequently, we draw attention to the limitations identified in the original approach and proceed to offer in-depth explanations of the proposed improved methodology.

3.1 *Original Sine Cosine Algorithm*

The optimization method known as the sine cosine algorithm (SCA), introduced by Mirjalili [26], is a popular approach utilized in the population-based tuning process. Typically, the optimization of these algorithms commences by initializing a set of

random solutions that are afterward iteratively assessed using a fitness function that incorporates a set of regulations, which constitute the fundamental framework of the tuning method.

In the exploration phase, the SCA employs a significant degree of randomness to rapidly examine diverse regions throughout the entire search realm. To accomplish this, random perturbations are introduced to the population. Nevertheless, as the exploitation phase follows, these arbitrary perturbations diminish, and gradual modifications are applied to the initial solutions. The search process of SCA is governed by Eq. (4).

$$X_i^{t+1} = \begin{cases} X_i^t + r_1 \times \sin(r_2) \times |r_3 P_i^t - X_i^t| & r_4 < 0.5 \\ X_i^t + r_1 \times \cos(r_2) \times |r_3 P_i^t - X_i^t| & r_4 \geq 0.5 \end{cases} \quad (4)$$

In Eq. (4), the symbol X_i^t represents the state of the observed individual for the i -th component during the t -th round of execution. The variables r_1 , r_2 , and r_3 denote random control parameters, whereas P_i corresponds to the position of the location factor in the i -th dimension. For further details regarding the SCA control parameters, refer to [26].

The inherent periodicity of the elementary trigonometric functions allows a solution to revolve around another individual, allowing easier exploitation of the space between this pair of individuals. To effectively examine the vast search region, solutions can additionally traverse outside the area defined by their respective locations. This is accomplished through the manipulation of the sequence of sine and cosine functions.

3.2 The Improved Sine Cosine Algorithm

While the SCA algorithm has proven to be a robust optimizer, like other metaheuristics approaches, it does possess certain limitations. Extensive simulations using benchmark functions have revealed that the elementary variant of the SCA algorithm exhibits a strong exploration capability. However, in subsequent iterations of the algorithm, it demonstrates a deficiency in exploitation power.

On the opposite, the firefly algorithm (FA) [39] is well-known for its powerful exploitation, given by the famous FA search Eq. (5):

$$x_i^{t+1} = x_i^t + \beta_0 \cdot e^{-\gamma r_{i,j}^2} (x_j^t - x_i^t) + \alpha^t (\kappa - 0.5) \quad (5)$$

The parameter responsible for randomization is represented by the symbol α , and it can be either a uniform distribution or a random number produced from a Gaussian distribution represented by κ . The distance among individuals i and j is indicated by $r_{i,j}$. Typically, the values of $[0, 1]$ have been commonly used for α , while the value of 1 has been widely employed as a suitable choice for β_0 to achieve significant

results across a majority of problems. For further details, please refer to the original FA publication [39].

The approach taken in this paper proposes the low-level hybridization of SCA and FA, defined as follows. During the first half of iterations (in the conducted experiments equal to 4, as the max count of iterations is 8), the algorithm performs SCA search according to the Eq. (4), to make use of the powerful SCA exploration. In the second half of iterations, however, when it is required to restrict the scope of the search and exploit promising areas, the algorithm alternates SCA and FA search (Eq. 5), as FA will help the exploitation in this particular case.

The improved method has been given the name HSCA. The pseudocode for the HSCA providing insights into inner workings is given in Algorithm 1.

Algorithm 1 Pseudocode of the HSCA algorithm

```

Produce a collection of starting solutions ( $X$ )
while  $t < \text{maxIter}$  do
  assess each solution with respect to its fitness function
  for every individual in ( $X$ ) do
    if  $t < \text{maxIter}/2$  then
      Execute SCA search as defined by the Eq. (4)
    else
      if  $t \% 2 = 0$  then
        Execute SCA search as defined by the Eq. (4)
      else
        Perform to the FA search as defined by Eq. (5)
  update algorithm's parameters
return The best established solution
  
```

4 Experiments and Discussion

The initial part of this section provides an introduction to the dataset and metrics utilized in the simulations, along with an outline of the fundamental simulation configuration, comparative analysis, and subsequent discussion.

4.1 Dataset

The dataset utilized for conducting classification simulations is accessible to general public, and was obtained from Kaggle.¹ It includes Maritime Mobile Service Identities (MMSIs) numbers, navigational status, speed over ground, course over ground, vessel heading, breadth, length, and draft, which are all retrieved from AIS data.

¹ <https://www.kaggle.com/datasets/eminskerkanerdonmez/ais-dataset>.

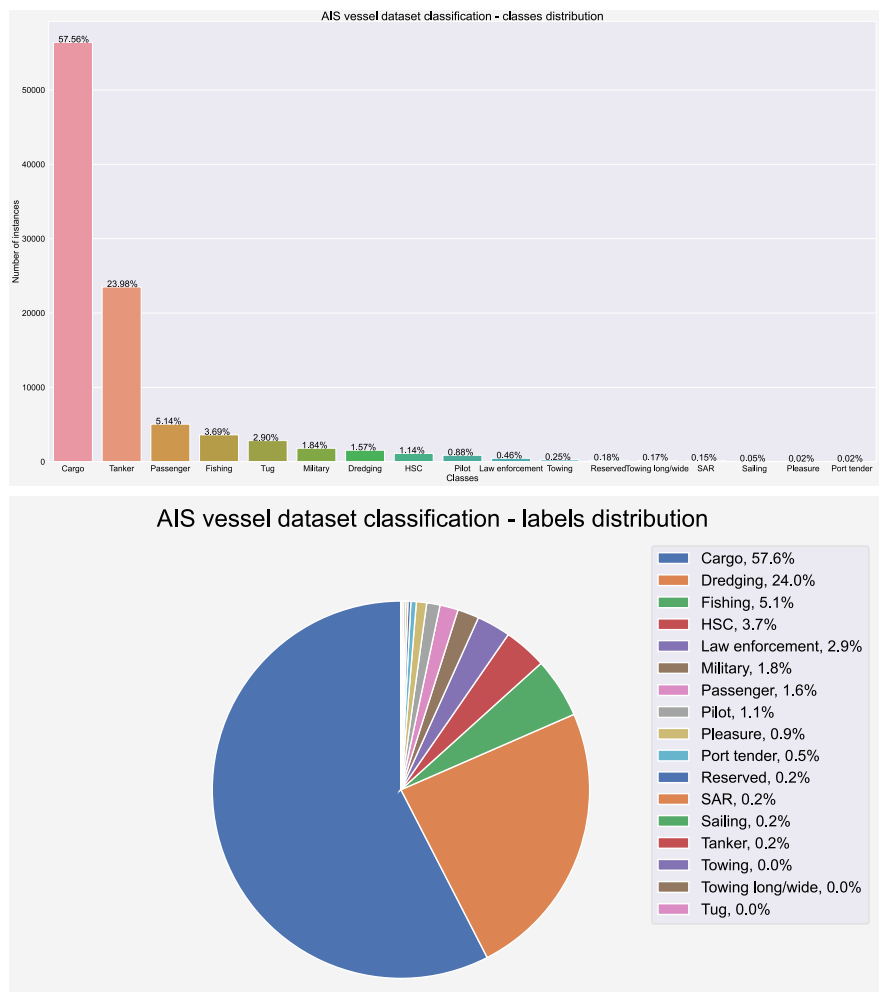


Fig. 1 Vessel classification dataset visualization

These characteristics were used to classify the kind of naval object. The navigation status attribute, which was originally supplied as a string, required conversion.

To enable the decision tree model’s capacity to operate with categorical characteristics in integer format, the text was first encoded as an integer. Nonetheless, this method produced less desired outcomes than one-hot encoding. As a result, the ship’s navigation state was ultimately encoded using one-hot encoding. The dataset is depicted in Fig. 1.

4.2 Experimental Setup

The AIS dataset described in the previous section was utilized to tune the XGBoost hyperparameters by applying the suggested HSCA algorithm. The hyperparameters that were subjected to optimization, accompanied by their respective search limits and variable types, are listed below:

- learning rate (η), domain: [0.1, 0.9], continual,
- *min_child_weight*, domain: [1, 10], continual,
- subsample, domain: [0.01, 1], continual,
- collsample_bytree, domain: [0.01, 1], continual,
- max_depth, domain: [3, 10], integer, and
- γ , domain: [0, 0.8], continual.

The aforementioned search boundaries were established by conducting practical experiments employing the trial and error method. To implement the proposed approach, the Python programming language was employed, making use of widely used Python libraries like scipy, numpy, and pandas. The XGBoost model that was used in this approach was acquired from the sci-kit learn package. The encoded solution vector is configured to have a length of six, matching the count of fine-tuned hyperparameters.

The dataset has been split in a random fashion, allocating 70% for training, and the rest 30% has been reserved for testing purposes.

The suggested hybrid XGBoost-HSCA model's performance has been evaluated by comparative analysis with six contending optimizers, that were namely an elementary version of SCA, genetic algorithm (GA) [29], PSO [22], whale optimization algorithm (WOA) [28], bat algorithm (BA) [40], and salp swarm algorithm (SSA) [27]. The experiments entailed executing each algorithm with the suggested control parameter values for the identical purpose—tuning XGBoost hyperparameters. All these contending methods were implemented and assessed separately by authors. A uniform configuration was adopted for all evaluated algorithms, comprising a population size of 8 individuals, 8 iterations in a single run, over a total of 30 independent executions.

The objective function used for maximization was Cohen's kappa coefficient κ [14], which measures the agreement between a pair of classification collections while accounting for classification randomness [25]. The calculation of κ follows the Eq. (6):

$$\kappa = \frac{p_o - p_e}{1 - p_e} = 1 - \frac{1 - p_o}{1 - p_e} \quad (6)$$

where p_o represents the observed values, while p_e depicts the expected values. It is important to note that the experiments also included monitoring the classification error.

4.3 Simulation Results and Discussion

This section showcases the experimental results achieved by the observed methods. Every table containing outcomes has the best score in each category noted in bold font. Table 1 yields the overall results with respect to the fitness function (Cohen’s kappa parameter in conducted experiments), while Table 2 shows the values of the classifying error indicator. The introduced XG-HSCA method has delivered the supreme scores for both the overall Cohen’s kappa results as well as the outcomes for the classification error. The suggested XG-HSCA was superior in all observed metrics, while XG-PSO attained the second best result, followed by the XG-SSA and XG-GA methods.

Table 3 exhibits the detailed metric values attained by the best-performance model optimized by HSCA algorithm. The aim was to get insight into the detailed per-class metrics with respect to precision, recall, and f1-score. One can note that XG-HSCA method exhibits high precision and recall for all classes, including excellent scores for minor classes as well. Looking at the overall accuracy, XG-HSCA achieved 0.997279, that was supreme in comparison to other contenting algorithms. Finally, the best-performing XGBoost set of hyperparameters, discovered by every regarded metaheuristics, has been shown in Table 4.

Table 1 Overall objective function (Cohen’s kappa) results

Method	Best	Worst	Mean	Median	Std	Var
XG-HSCA	0.995504	0.994771	0.995101	0.995026	0.000275	7.54E−08
XG-SCA	0.993592	0.984966	0.989528	0.990753	0.003087	9.53E−06
XG-GA	0.994041	0.969274	0.986571	0.989701	0.008458	7.15E−05
XG-PSO	0.994322	0.983340	0.990345	0.992298	0.004068	1.65E−05
XG-BA	0.993702	0.991678	0.993047	0.993424	0.000722	5.21E−07
XG-WOA	0.993422	0.987625	0.991041	0.991229	0.001943	3.77E−06
XG-SSA	0.994044	0.980190	0.989461	0.990408	0.004610	2.13E−05

Table 2 Overall indicator function (classification error) outcomes

Method	Best	Worst	Mean	Median	Std	Var
XG-HSCA	0.002721	0.003164	0.002965	0.003011	0.000166	2.75E−08
XG-SCA	0.003878	0.009083	0.006333	0.005596	0.001863	3.47E−06
XG-GA	0.003606	0.018539	0.008113	0.006225	0.005100	2.60E−05
XG-PSO	0.003436	0.010069	0.005840	0.004660	0.002457	6.04E−06
XG-BA	0.003810	0.005035	0.004207	0.003980	0.000436	1.90E−07
XG-WOA	0.003980	0.007484	0.005420	0.005307	0.001174	1.38E−06
XG-SSA	0.003606	0.011974	0.006373	0.005800	0.002785	7.76E−06

Table 3 Detailed per-class metrics of the top performing XG-HSCA tuned model

Class	Precision	Recall	f_1 -score	Support
0	0.998582	0.999173	0.998877	16,921
1	0.984848	0.986985	0.985915	461
2	1.000000	0.995383	0.997686	1083
3	0.988095	0.991045	0.989568	335
4	1.000000	1.000000	1.000000	136
5	0.992606	0.992606	0.992606	541
6	0.999337	0.997354	0.998345	1512
7	0.996154	1.000000	0.998073	259
8	1.000000	1.000000	1.000000	7
9	1.000000	1.000000	1.000000	5
10	0.901961	0.867925	0.884615	53
11	1.000000	0.955556	0.977273	45
12	0.769231	0.714286	0.740741	14
13	0.998015	0.998440	0.998227	7050
14	0.929577	0.904110	0.916667	73
15	0.857143	0.960000	0.905660	50
16	0.995266	0.987089	0.991161	852
Accuracy	0.997279	0.997279	0.997279	0.997279
Macro avg	0.965342	0.961762	0.963260	29397
Weighted avg	0.997289	0.997279	0.997276	29397

Table 4 Top-performing models' determined hyperparameters values for multiclass classification

Method	Learning rate	Min child weight	Subsample	Colsample by tree	Max depth	Gamma
XG-HSCA	0.766393	1.000000	1.000000	0.851902	10	0.269730
XG-SCA	0.578478	1.000000	1.000000	1.000000	10	0.452535
XG-GA	0.715723	1.000000	0.751756	1.000000	10	0.508815
XG-PSO	0.812949	1.000000	0.865553	0.982934	10	0.019729
XG-BA	0.621127	1.000000	1.000000	1.000000	10	0.788611
XG-WOA	0.785501	6.423267	1.000000	1.000000	10	0.800000
XG-SSA	0.900000	1.126151	0.857129	0.724793	10	0.043916

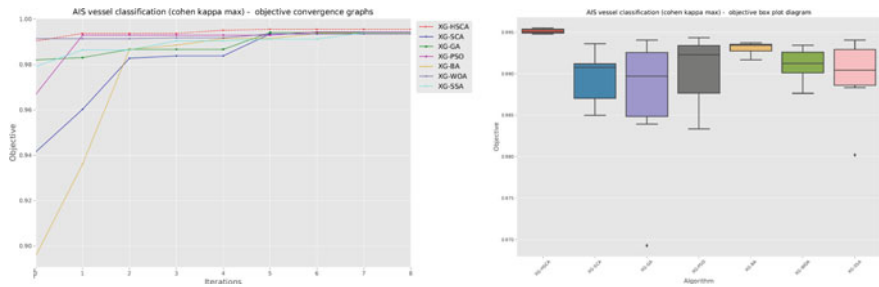


Fig. 2 Convergence graphs and box-plot diagrams of the observed methods for the fitness (Cohen’s kappa)

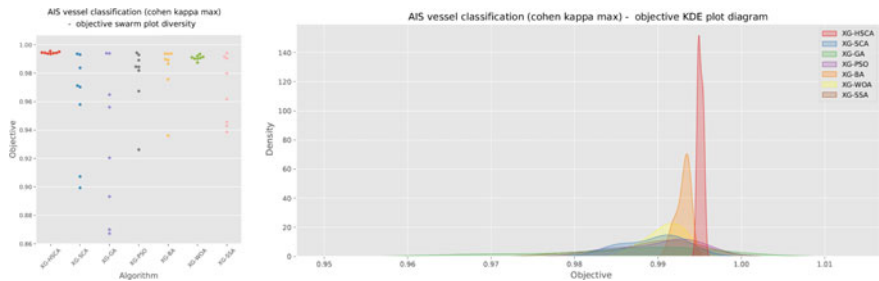


Fig. 3 Swarm and KDE diagrams of the observed methods for the fitness (Cohen’s kappa)

To visualize the performance of the regarded approaches, Fig. 2 presents convergence graphs illustrating the convergence speed of the proposed HSCA, which demonstrates a rapid convergence. Furthermore, box-plots displayed in Fig. 2 suggest the consistent and stable results of the suggested method over multiple independent executions. Moreover, Fig. 3 showcases KDE diagrams and swarm plots. The swarm plots provide insight into the diversity of the population during the final iteration for the best run of every regarded metaheuristics. They reveal that all solutions generated by the proposed HSCA fall in close proximity to the best score.

Last but not least, aiming to exhibit the outcomes of the proposed HSCA on the regarded AIS dataset, Fig. 4 showcases the confusion matrix and PR curve diagram.

Considering all factors, the XG-HSCA method introduced in this study exhibited remarkable performance on the AIS dataset under consideration. It is worth mentioning that other XGBoost models fine-tuned by other powerful contemporary metaheuristic algorithms have also attained commendable results. In comparison to other basic metaheuristics included in the comparisons, the elementary variant of SCA also demonstrated decent performance, but was significantly outperformed by devised HSCA algorithm.

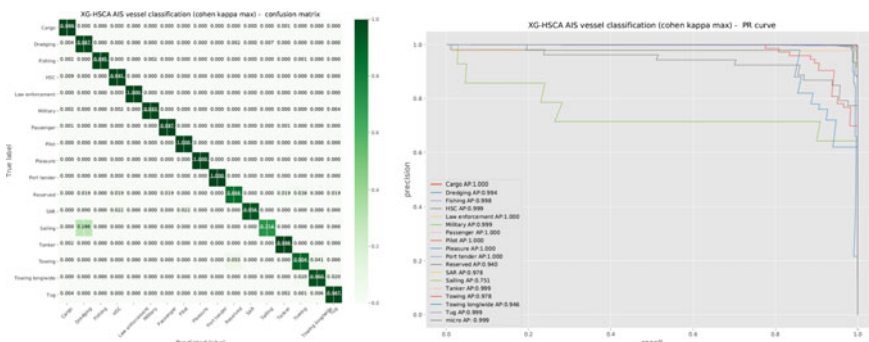


Fig. 4 Confusion matrix and PR curve attained by the suggested XG-HSCA

5 Conclusion

The exploration of machine learning methods, especially those fine-tuned using the devised HSCA algorithm, has yielded valuable insights into addressing the important task of ship classification.

The created models in this study have demonstrated exceptional level of performance in accurately identifying vessel types, showcasing the effectiveness and potential of combining machine learning with proposed HSCA metaheuristics. Particularly notable is the suggested XG-HSCA model, that achieved an impressive accuracy of 0.997279 in the vessel classification task, highlighting its ability to tackle complex classification challenges commonly seen in real-world scenarios.

The simulation outcomes emphasize the importance and effectiveness of applying advanced and tuned machine learning approaches for the purpose of vessel identification and classification. Nevertheless, despite the encouraging outcomes, there remains significant unexplored potential in this field. It is necessary that future research focus even deeper to exploring and enhancing the utilization of machine learning algorithms in various classification problems across different domains, including cybersecurity, healthcare, cloud systems, and other contemporary technologies.

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